

Multiple Choice (10 marks)

1. A group of students observed the formation of bubbles when water boiled. What happened to matter because of the physical change to the water?
 - (a) New matter was created in the bubbles.
 - (b) Old matter was destroyed as the water turned to gas.
 - (c) Matter was conserved when it changed form.
 - (d) The composition of the matter changed.
2. Why does sexual reproduction result in more genetic diversity than asexual reproduction?
 - (a) Traits from two parents are combined.
 - (b) More organisms reproduce this way.
 - (c) Offspring grow in different environments.
 - (d) Offspring come from identical parents.
3. Paramecium and volvox can both reproduce sexually. How is sexual reproduction in volvox different than sexual reproduction in paramecium?
 - (a) Volvox colonies produce gametes.
 - (b) Volvox cells undergo conjugation.
 - (c) Volvox colonies produce single-celled offspring.
 - (d) Volvox cells reproduce by vegetative budding.
4. The human population is increasing every day. How does this increase in population affect the natural resources that maintain humans?
 - (a) increases the supply of resources
 - (b) decreases the amount of resources
 - (c) results in less demand for resources
 - (d) improves the quality of the resources
5. All of these are examples of the ways Earth and the Moon interact except
 - (a) the phases of the Moon.
 - (b) the tides on Earth.
 - (c) seasons on Earth.
 - (d) lunar eclipses.

True / False (10 marks)

Answer True or False

6. True or False: Hawks have sharp claws called talons, which are an inherited trait that helps them capture prey.
7. True or False: A student should notify the teacher first if a container tips over and spills liquid during a laboratory exercise.
8. True or False: The new organism could belong to the kingdoms of Protists and Plants due to its multicellular and autotrophic nature.
9. True or False: Pencils falling from the same height effectively demonstrate the effects of gravity on moving objects.
10. True or False: If the data do not support the hypothesis, it is important to check for errors and repeat the experiment.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain the process by which a plastic rod becomes negatively charged when rubbed with fur, including the role of electrons and the concept of static electricity.
12. Discuss how communication within cells, particularly the role of special molecules that transmit messages from the membrane to the nucleus, is similar to processes in a specific body system.

End of Paper